1. **-- farmer, goat, grass, wolf = L, L, L, L**
2. **S**
3. **-- farmer , goat , grass , wolf = R, R, R, R**

We have to compose S as a sequence of assignment statements such that in none of the intermediate states

1. goat and wolf have the same value but farmer has the opposite value, or
2. goat and grass have the same value but farmer has the opposite value.Subject to these constraints, a sequence of assignments and the state after each assignment are shown in Figure 7.3.

1. -- farmer, goat, grass, wolf = L, L, L, L
2. **farmer, goat := R, R**
3. -- farmer , goat , grass , wolf = R, R, L, L
4. **farmer := L**
5. farmer, goat, grass, wolf = L, R, L, L
6. **farmer, grass := R, R**
7. -- farmer , goat , grass , wolf = R, R, R, L
8. **farmer, goat := L, L**
9. -- farmer, goat, grass, wolf = L, L, R, L
10. **farmer, wolf := R, R**
11. -- farmer , goat , grass , wolf = R, L, R, R
12. **farmer : = L**
13. -- farmer , goat , grass , wolf = L, L, R, R
14. **farmer , goat : = R, R**
15. -- farmer , goat , grass , wolf = R, R, R, R

Figure 7.3: Sequence of assignments for goat, grass and wolf problem

Other than lines (1) and (15), in line (7), goat and grass have the same value, but farmer also has the same value as they. In line (9), goat and wolf have the same value, but farmer also has the same value as they. Thus, the sequence has achieved the goal state, without violating the constraints.

7.2.2 Alternative statement

A condition is a phrase that describes a test of the state. If C is a condition and both

S1 and S2 are statements, then

```
if  C
    S1
else
    S2
```

is a statement, called an alternative statement, that describes the following action:

1. Test whether C is true or false.
2. If C is true, then do S1; otherwise do S2.

In pseudo code, the two alternatives S1 and S2 are indicated by indenting them from the keywords if and else, respectively. Alternative control flow is depicted in the flowchart of Figure 2.4. Condition C has two outgoing arrows, labeled true and false. The true arrow points to the S1 box. The false arrow points to the S2 box. Out going arrows of S1 and S2 point to the same box, the box after the alternative statement.

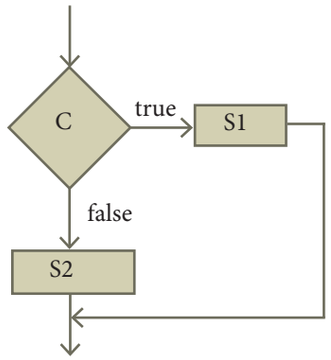


Figure 7.4: Alternative control flow

Chapter 7 Page 088-101.indd 91 3/24/2020 9:14:50 AM